



Let S be a set, and let P and Q be predicates on S . Prove the proposition

$$\forall_{x \in S} (P(x) \Rightarrow Q(x)) \Rightarrow (\forall_{y \in S} (P(y)) \Rightarrow \forall_{z \in S} (Q(z)))$$

twice, in flag format.

- (a) In natural deduction. Name the rule used on every line.
- (b) In λP . In the context $\Gamma \equiv S : *, P : S \rightarrow *, Q : S \rightarrow *$, find a term whose type codes the proposition, and give its derivation. Use the same flags and the same line numbers as in (a).
- (c) Which rule of λP did each rule of (a) become?

Nederpelt & Geuvers, *Type Theory and Formal Proof* (CUP, 2014), Exercise 5.9 (b).

I Definitions

Flag format. A flag holds an assumption or a declaration, and the lines under its pole may use it. Each line names its rule and the lines or flags it uses. A declaration $x : A$ is a flag too, and (*abst*) closes it.

Natural deduction. An elimination rule uses an operation, an introduction rule proves it and closes its flag.

$$\begin{array}{c} \frac{A \Rightarrow B \quad A}{B} (\Rightarrow\text{-elim}) \qquad \frac{\text{Assume } A \cdots B}{A \Rightarrow B} (\Rightarrow\text{-intro}) \\ \frac{\forall_{x \in S} (P(x)) \quad N \in S}{P(N)} (\forall\text{-elim}) \qquad \frac{\text{Let } x \in S \cdots P(x)}{\forall_{x \in S} (P(x))} (\forall\text{-intro}) \end{array}$$

Propositions as types. A term of a type is a proof of the proposition that the type codes:

in logic	in λP	in logic	in λP	in logic	in λP
S is a set	$S : *$	A is a proposition	$A : *$	P is a predicate on S	$P : S \rightarrow *$
$a \in S$	$a : S$	p proves A	$p : A$	$P(x)$	Px
		$A \Rightarrow B$	$A \rightarrow B$	$\forall_{x \in S} (P(x))$	$\Pi x : S. Px$

$*$ is the type of all types. $\Pi x : A. B$ is the type of the functions sending each $x : A$ to a term of type B , where B may mention x ; $A \rightarrow B$ is the case in which it does not.

The two rules of λP you need. $B[x := N]$ is B with N substituted for x .

$$\frac{\Gamma \vdash M : \Pi x : A. B \quad \Gamma \vdash N : A}{\Gamma \vdash MN : B[x := N]} (\text{appl}) \qquad \frac{\Gamma, x : A \vdash M : B}{\Gamma \vdash \lambda x : A. M : \Pi x : A. B} (\text{abst})$$

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Answer

Out of room? Continue on the back of the page.